

Chapter 4

Fourier Series

Learning Objectives

By the end of this chapter, students should be able to

1. define periodic functions;
2. describe the symmetric properties of even and odd periodic functions;
3. find the Fourier series of a periodic function.

The French mathematician *Joseph Fourier* (1768 – 1830) first used an infinite series of sine and cosine functions to solve a long-standing problem on heat transfer. This trigonometric series, later known as Fourier series, was found to fill in the gap for approximating functions where the Taylor (and Maclaurin) series fail. The Taylor and Maclaurin series approximate function values very well near a point, while the Fourier series is well-suited for approximating functions on wide intervals, particularly functions with repeating cycles, i.e. periodic functions. This is useful because periodic functions abound in science and engineering, such as heartbeat, tides, vibrating strings, radio signals and alternating currents, etc.

We begin with Section 4.1 by defining the periodic function. The Fourier series formula follows immediately in Section 4.2, where we work through a few typical examples. Section 4.3 is devoted to the Fourier series of even/odd functions; their Fourier series take simpler forms. A simple application of Fourier series is presented in Section 4.4 and exercises are given in Section 4.5 for further practice.

4.1 Periodic Functions

Definition 4.1.1 (Periodic Function)

A function $f(t)$ is a **periodic function** if there is a positive number T such that

$$f(t) = f(t + T)$$

for all t in the domain of f . The **period** of f is the smallest such T .

A periodic function with period T is referred to as T -periodic function.

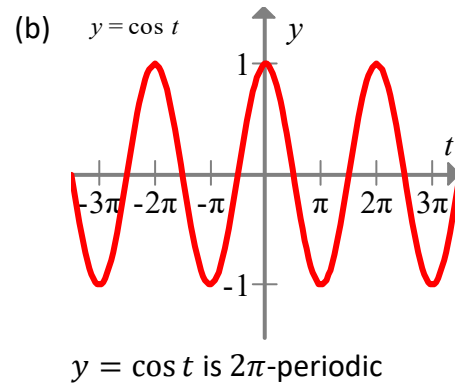
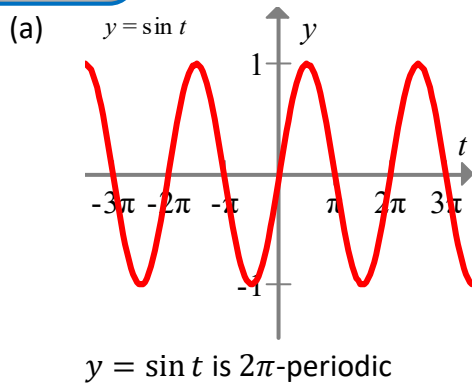
Example 4.1.2

Sketch the following functions and determine their periods:

(a) $y = \sin t$

(b) $y = \cos t$

Solution



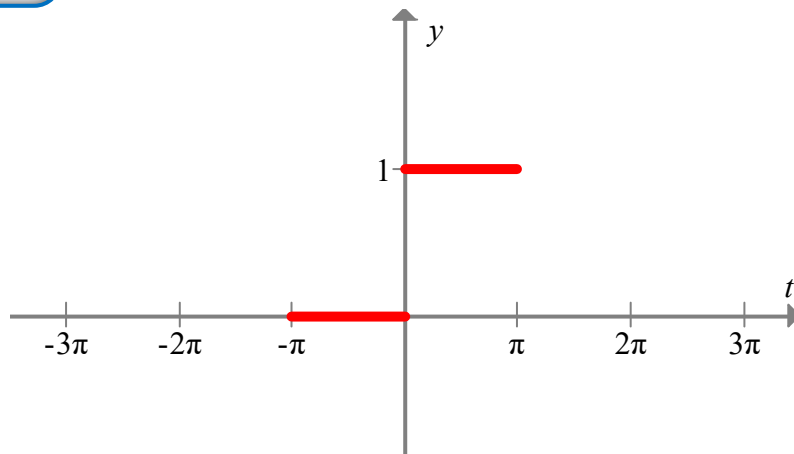
A function $f(t)$ defined on the interval $-L < t < L$ can be extended to a $2L$ -periodic function by imposing the condition $f(t) = f(t + 2L)$ for all t , as shown in Examples 4.1.3 and 4.1.4.

Example 4.1.3

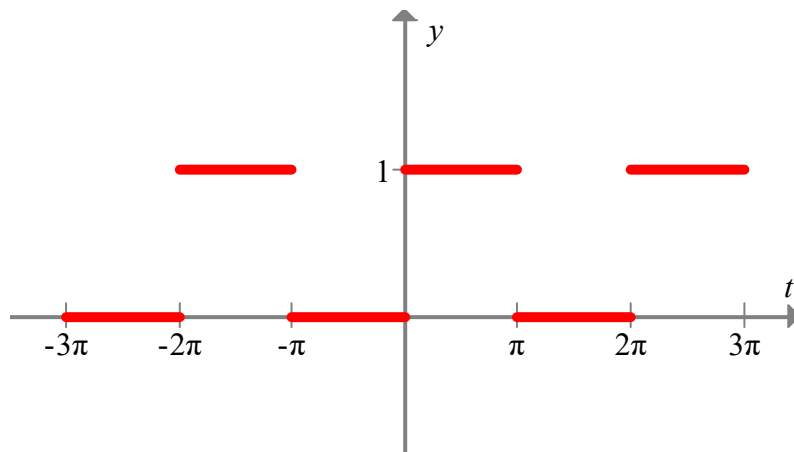
- (a) Sketch the graph of $f(t) = \begin{cases} 0 & \text{for } -\pi < t < 0 \\ 1 & \text{for } 0 < t < \pi \end{cases}$ on the interval $-\pi < t < \pi$.
- (b) Hence, sketch the periodic extension $f(t) = f(t + 2\pi)$ for all t on the interval $-3\pi < t < 3\pi$.

Solution

(a)



(b)



Example 4.1.4

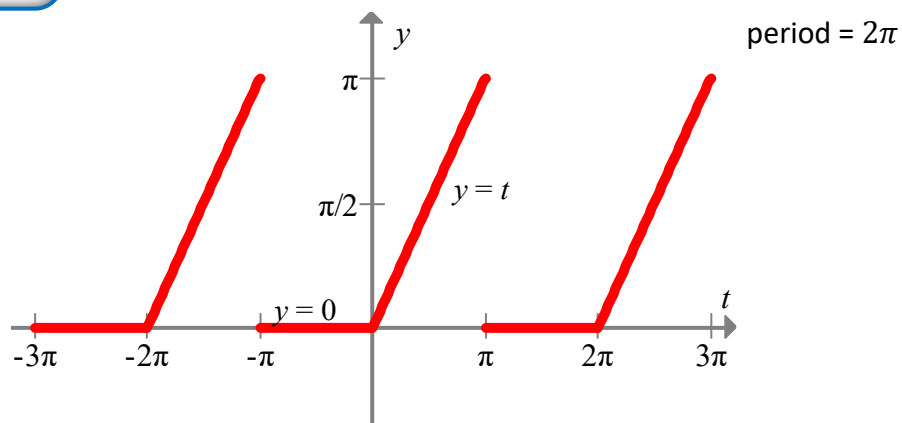
Sketch three cycles of the graph of the following periodic functions and determine their periods.

$$(a) \quad f(t) = \begin{cases} 0 & \text{for } -\pi < t < 0 \\ t & \text{for } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi)$$

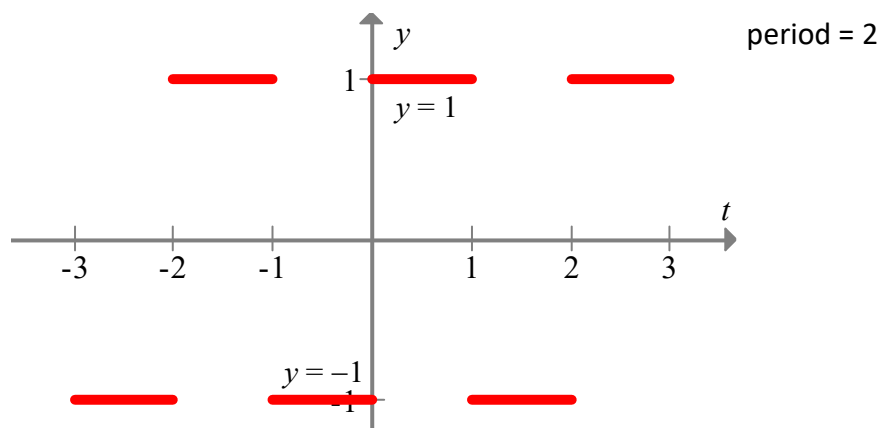
$$(b) \quad g(t) = \begin{cases} -1 & \text{for } -1 < t < 0 \\ 1 & \text{for } 0 < t < 1 \end{cases} \quad \text{and} \quad g(t) = g(t + 2)$$

Solution

(a)



(b)



4.2 The Fourier Series

The Fourier series is an infinite series of sine and cosine functions, and is suitable for approximating periodic functions.

Definition 4.2.1 (Fourier Series)

Let $f(t)$ be a T -periodic function. The Fourier series of $f(t)$ is an expansion of $f(t)$ in the form of the trigonometric series

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{2n\pi t}{T}\right) + b_n \sin\left(\frac{2n\pi t}{T}\right) \right\}$$

where a_0 , a_n , and b_n are the **Fourier coefficients** given by

$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos\left(\frac{2n\pi t}{T}\right) dt \text{ for } n = 1, 2, 3, \dots$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin\left(\frac{2n\pi t}{T}\right) dt \text{ for } n = 1, 2, 3, \dots$$

Remark. As an infinite series, the convergence of the Fourier series is a very important concern. However, discussion on this convergence is beyond the scope of this text as it involves very advanced mathematics. We shall assume convergence of the Fourier series for all points except at discontinuous points.

These results are often used when computing Fourier series.

$$\sin(n\pi) = 0 \quad \text{and} \quad \cos(n\pi) = (-1)^n \quad \text{for all integers } n$$

When calculating the Fourier series of a periodic function, the following procedure is a good guide.

Procedure to Calculate the Fourier Series of a Periodic Function

Step 1. Identify the period T .

Step 2. Calculate the Fourier coefficients a_0, a_n, b_n .

Step 3. Write the Fourier series of f in sigma notation (see Definition 4.2.1).

Step 4. Where possible, simplify the coefficients.

In Step 4 of the procedure above, the Fourier coefficients a_n or b_n can be simplified if they contain either the factor of $(1 - (-1)^n)$ or $(1 + (-1)^n)$. This is done by observing its alternating values as follows.

$$1 - (-1)^n = \begin{cases} 2 & \text{for } n = 2k - 1 \quad \text{i.e. } n \text{ is odd} \\ 0 & \text{for } n = 2k \quad \text{i.e. } n \text{ is even} \end{cases}$$

$$1 + (-1)^n = \begin{cases} 0 & \text{for } n = 2k - 1 \quad \text{i.e. } n \text{ is odd} \\ 2 & \text{for } n = 2k \quad \text{i.e. } n \text{ is even} \end{cases}$$

Example 4.2.2

$$(a) \quad \sum_{n=1}^{\infty} \left(\frac{1 - (-1)^n}{n\pi} \right) \sin(n\pi t) = \sum_{k=1}^{\infty} \frac{2}{(2k-1)\pi} \sin((2k-1)\pi t)$$

$$(b) \quad \sum_{n=1}^{\infty} \left(\frac{1 + (-1)^n}{n\pi} \right) \sin(n\pi t) = \sum_{k=1}^{\infty} \frac{2}{2k\pi} \sin(2k\pi t)$$

Example 4.2.3

Calculate the Fourier series of $f(t)$ given by

$$f(t) = \begin{cases} 0 & \text{for } -1 < t < 0 \\ 1 & \text{for } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2).$$

Solution

Step 1. $f(t) = f(t + 2) \Rightarrow T = 2$

$$\text{Step 2. } a_0 = \frac{2}{2} \int_{-1}^1 f(t) dt = 0 + \int_0^1 1 dt = 1.$$

$$\begin{aligned} a_n &= \frac{2}{2} \int_{-1}^1 f(t) \cos\left(\frac{2n\pi t}{2}\right) dt = 0 + \int_0^1 1 \cdot \cos(n\pi t) dt \\ &= \left[\frac{\sin(n\pi t)}{n\pi} \right]_0^1 = \frac{\sin(n\pi)}{n\pi} - \frac{\sin(0)}{n\pi} = 0. \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{2} \int_{-1}^1 f(t) \sin\left(\frac{2n\pi t}{2}\right) dt = 0 + \int_0^1 1 \cdot \sin(n\pi t) dt \\ &= \left[-\frac{\cos(n\pi t)}{n\pi} \right]_0^1 = -\frac{\cos(n\pi)}{n\pi} + \frac{\cos(0)}{n\pi} = \frac{1 - (-1)^n}{n\pi}. \end{aligned}$$

Step 3. The Fourier series of $f(t)$ is $f(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{1 - (-1)^n}{n\pi} \right) \sin(n\pi t)$.

Step 4. $f(t) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{(2k-1)\pi} \sin((2k-1)\pi t)$, cf. Example 4.2.2(a).

Let us consider the Fourier series of the function $f(t)$ in Example 4.2.3. Denoting the n -th partial sum of the Fourier series as

$$S_n(t) = \frac{1}{2} + \sum_{k=1}^n \frac{2}{(2k-1)\pi} \sin((2k-1)\pi t),$$

we can see graphically how $S_n(t)$ approximates $f(t)$ as n gets larger, as shown in Figure 4.1 below.

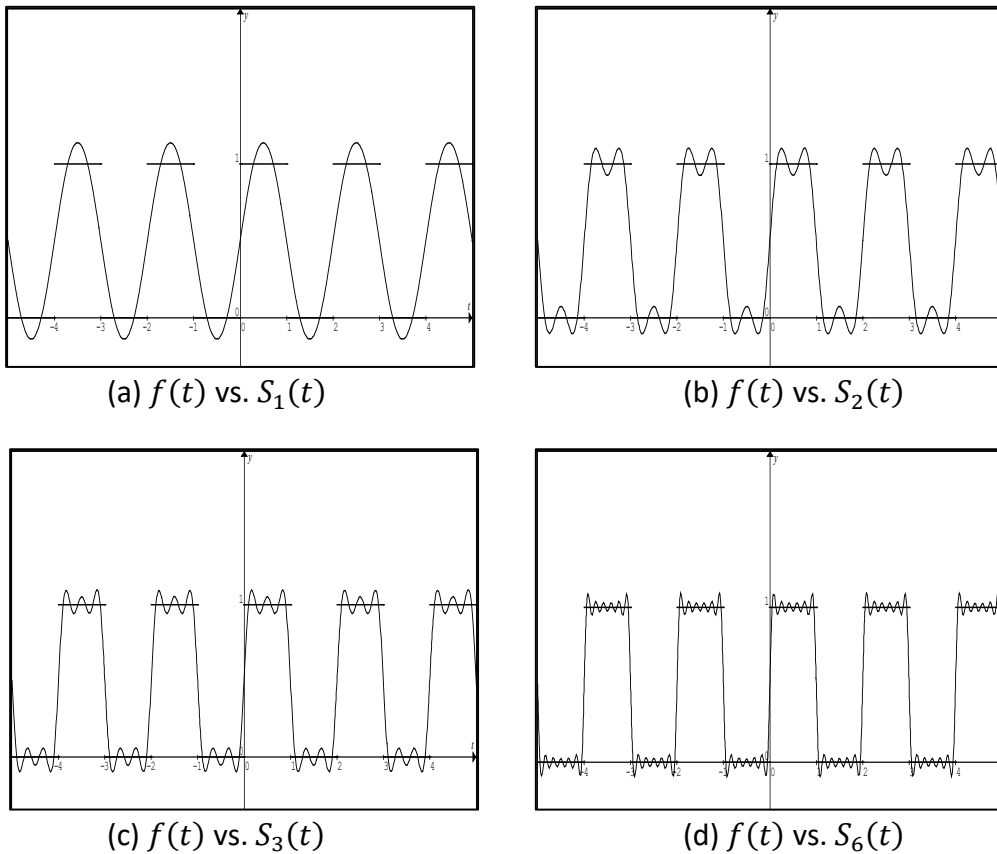


Figure 4.1. Graphs of $f(t)$ versus $S_n(t)$

We observe from the graphs in Figure 4.1 above that as n gets larger, the graphs of $S_n(t)$ resemble that of $f(t)$ more and more.

Example 4.2.4

Calculate the Fourier series of $f(t)$ given by

$$f(t) = \begin{cases} 0 & \text{for } -\pi < t < 0 \\ t & \text{for } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi).$$

Solution

Step 1. $f(t) = f(t + 2\pi) \Rightarrow T = 2\pi$.

$$\text{Step 2. } a_0 = \frac{2}{2\pi} \int_{-\pi}^{\pi} f(t) dt = \frac{1}{\pi} \left[0 + \int_0^{\pi} t dt \right] = \frac{1}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} = \frac{\pi}{2}.$$

$$\begin{aligned} a_n &= \frac{2}{2\pi} \int_{-\pi}^{\pi} f(t) \cos\left(\frac{2n\pi t}{2\pi}\right) dt = \frac{1}{\pi} \left[0 + \int_0^{\pi} t \cdot \cos(nt) dt \right] \\ &= \frac{1}{\pi} \left[\frac{t \sin(nt)}{n} + \frac{\cos(nt)}{n^2} \right]_0^{\pi} = \frac{1}{\pi} \left[\frac{\pi \sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2} \right] - \frac{1}{\pi} \left[0 + \frac{\cos 0}{n^2} \right] \\ &= \frac{1}{\pi} \left[0 + \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] = \frac{(-1)^n - 1}{\pi n^2}. \end{aligned}$$

$$\text{Similarly, } b_n = \frac{-(-1)^n}{n}.$$

Step 3. The Fourier series of $f(t)$ is

$$f(t) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left\{ \left(\frac{(-1)^n - 1}{\pi n^2} \right) \cos(nt) - \frac{(-1)^n}{n} \sin(nt) \right\}$$

$$\text{Step 4. } f(t) = \frac{\pi}{4} - \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\cos(2k-1)t}{(2k-1)^2} - \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nt).$$

4.3 Fourier series of Even/Odd Functions

The computations of the Fourier series coefficients of even or odd functions are simpler. Thus, an effort to determine whether a function is even or odd helps make computations faster. We will first define what odd and even functions are and then look at some of their properties.

Definition 4.3.1 (Even and Odd Functions)

A function f is said to be **even** if $f(-t) = f(t)$ for all t in the domain of f .

A function f is said to be **odd** if $f(-t) = -f(t)$ for all t in the domain of f .

Remark. The graph of an even function is *symmetrical about the vertical axis*, while the graph of an odd function is *symmetrical about the origin*. There are also functions that are neither even nor odd.

Example 4.3.2

(a) The function $h(t)$ given by

$$h(t) = |t| \text{ for } -1 < t < 1 \text{ and } h(t) = h(t + 2)$$

is an even function. Its graph is symmetric about the vertical axis in Figure 4.2.

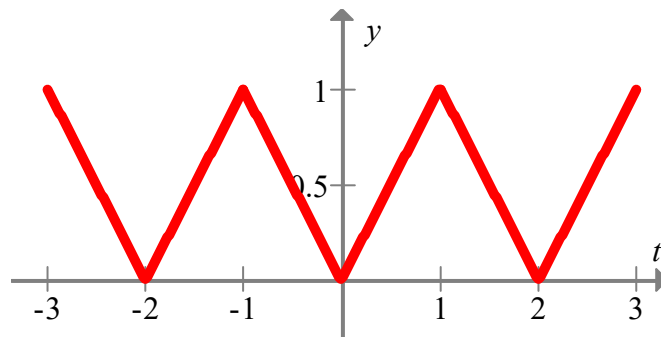


Figure 4.2. Graph of $h(t)$

- (b) The function $y(t)$ given by

$$y(t) = t \text{ for } -\pi < t < \pi \quad \text{and} \quad y(t) = y(t + 2\pi)$$

is an odd function. Its graph is symmetric about the origin in Figure 4.3.

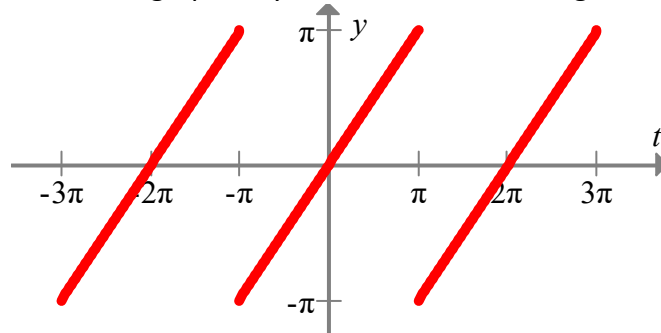


Figure 4.3. Graph of $y(t)$

- (c) The function $f(t)$ defined in Example 4.1.4(a) is neither even nor odd. Observe that its graph is not symmetrical about the vertical axis, neither is it symmetrical about the origin.
- (d) $y = \sin t$ is an odd function and $y = \cos t$ is an even function, cf. Example 4.1.2. In general, $y = \sin nt$ is an odd function and $y = \cos nt$ is an even function for any non-zero number n .

We will use graphical observation as in Example 4.3.2 to conclude whether a given function is even, odd or neither. For our purpose, we do not require a formal proof based on Definition 4.3.1.

The symmetries of even/odd functions give rise to the following properties.

Theorem 4.3.3 (Products of Even/Odd Functions)

- (a) The product of two even functions is even.
- (b) The product of two odd functions is even.
- (c) The product of an even function and an odd function is odd.

Theorem 4.4.3 is analogous to the product of positive/negative numbers, i.e. EVEN function $\sim$ POSITIVE number, and ODD function $\sim$ NEGATIVE number.

Theorem 4.3.4 (Integral of Even/Odd Functions)

Let $L > 0$.

(a) If $g(t)$ is an even function, then
$$\int_{-L}^L g(t) dt = 2 \int_0^L g(t) dt .$$

(b) If $g(t)$ is an odd function, then
$$\int_{-L}^L g(t) dt = 0 .$$

Proof

$$\begin{aligned} \text{(a)} \quad \int_{-L}^L g(t) dt &= \int_{-L}^0 g(t) dt + \int_0^L g(t) dt = - \int_L^0 g(-u) du + \int_0^L g(t) dt \\ &= \int_0^L g(u) du + \int_0^L g(t) dt = 2 \int_0^L g(t) dt \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_{-L}^L g(t) dt &= \int_{-L}^0 g(t) dt + \int_0^L g(t) dt = - \int_L^0 g(-u) du + \int_0^L g(t) dt \\ &= - \int_0^L g(u) du + \int_0^L g(t) dt = 0 \end{aligned}$$

Example 4.3.5

(a) $y = \sin t \cos t$ is an odd function, and $\int_{-\pi}^{\pi} \sin t \cos t \, dt = 0$.

(b) $y = \sin t \sin nt$ is an even function, and $\int_{-\pi}^{\pi} \sin t \cos nt \, dt = 2 \int_0^{\pi} \sin t \cos nt \, dt$.

(c) $y = t \cos nt$ is an odd function, and $\int_{-\pi}^{\pi} t \cos nt \, dt = 0$.

(d) $y = t^2 \cos nt$ is an even function, and $\int_{-\pi}^{\pi} t^2 \cos nt \, dt = 2 \int_0^{\pi} t^2 \cos nt \, dt$.

The following is a result from Theorem 4.3.4.

Corollary 4.3.6 (Fourier Coefficients of Even/Odd Functions)

Let $f(t)$ be a T -periodic function with Fourier coefficients a_0, a_n, b_n .

(a) If $f(t)$ is an even function, then $b_n = 0$ for all positive integers n .

(b) If $f(t)$ is an odd function, then $a_0 = a_n = 0$ for all positive integers n .

Proof

From Theorem 4.3.4, it is sufficient to show that for each case, the corresponding Fourier coefficient has odd function integrand.

(a) $f(t)$ is an even function $\Rightarrow f(t) \sin\left(\frac{2n\pi t}{T}\right)$ is an odd function.

(b) $f(t)$ is an odd function $\Rightarrow f(t) \cos\left(\frac{2n\pi t}{T}\right)$ is an odd function.

Example 4.3.7

Calculate the Fourier series of $f(t)$ given by

$$f(t) = \begin{cases} -1 & \text{if } -\pi < t < 0 \\ 1 & \text{if } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi).$$

Solution

Step 1. $f(t) = f(t + 2\pi) \Rightarrow T = 2\pi$.

Step 2. The graph of $f(t)$ is symmetrical about the origin (cf. Example 4.1.4(b)), so $f(t)$ is an odd function, and hence, $a_0 = a_n = 0$ for all positive integers of n .

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt \\ &= \frac{2}{\pi} \int_0^{\pi} f(t) \sin(nt) dt \quad \text{since } f(t) \sin(nt) \text{ is an even function} \\ &= \frac{2}{\pi} \int_0^{\pi} 1 \cdot \sin(nt) dt = \frac{2}{\pi} \left[\frac{-\cos(nt)}{n} \right]_0^{\pi} = \frac{2}{\pi} \left(\frac{-\cos(n\pi) + \cos 0}{n} \right) \\ &= \frac{2}{\pi} \left(\frac{1 - (-1)^n}{n} \right) \quad \text{for all positive integers } n. \end{aligned}$$

Step 3. The Fourier series of $f(t)$ is $f(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left(\frac{1 - (-1)^n}{n} \right) \sin(nt)$.

Step 4.
$$f(t) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin((2k-1)t)}{2k-1}$$

Example 4.3.8

Calculate the Fourier series of **Sawtooth function** $f(t)$ given by

$$f(t) = t \text{ for } -\pi < t < \pi \text{ and } f(t) = f(t + 2\pi).$$

Solution

Step 1. $f(t) = f(t + 2\pi) \Rightarrow T = 2\pi$.

Step 2. The graph of $f(t)$ is symmetrical about the origin (cf. Example 4.3.2(b)), so $f(t)$ is an odd function, and hence, $a_0 = a_n = 0$ for all positive integers of n .

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt \\ &= \frac{2}{\pi} \int_0^{\pi} f(t) \sin(nt) dt \quad \text{since } f(t) \sin(nt) \text{ is an even function} \\ &= \frac{2}{\pi} \int_0^{\pi} t \sin(nt) dt = \frac{2}{\pi} \left[-\frac{t \cos(nt)}{n} + \frac{\sin(nt)}{n^2} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\left(-\frac{\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} \right) - \left(0 + \frac{\sin 0}{n^2} \right) \right] \\ &= \frac{2}{\pi} \left[\left(-\frac{\pi(-1)^n}{n} + \frac{0}{n} \right) \right] = \frac{2(-1)^{n+1}}{n} \text{ for all positive integers } n. \end{aligned}$$

Step 3. The Fourier series of $f(t)$ is $f(t) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nt)$.

Example 4.3.9

Calculate the Fourier series of $f(t)$ given by

$$f(t) = |t| \text{ for } -1 < t < 1 \text{ and } f(t) = f(t + 2).$$

Solution

Step 1. $f(t) = f(t + 2) \Rightarrow T = 2$.

Step 2. The graph of $f(t)$ is symmetrical about the vertical axis (cf. Example 4.3.2(a)), so $f(t)$ is an even function, and hence, $b_n = 0$ for all positive integers of n .

$$a_0 = \int_{-1}^1 f(t) dt = 2 \int_0^1 f(t) dt = 2 \int_0^1 t dt = \left[t^2 \right]_0^1 = 1$$

$$\begin{aligned} a_n &= \int_{-1}^1 f(t) \cos(n\pi t) dt = 2 \int_0^1 f(t) \cos(n\pi t) dt \\ &= 2 \int_0^1 t \cos(n\pi t) dt = 2 \left[\frac{t \sin(n\pi t)}{n\pi} + \frac{\cos(n\pi t)}{(n\pi)^2} \right]_0^1 \\ &= 2 \left(\frac{(-1)^n - 1}{(n\pi)^2} \right) \text{ for all positive integers } n. \end{aligned}$$

Step 3. The Fourier series of $f(t)$ is

$$f(t) = \frac{1}{2} + 2 \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{(n\pi)^2} \right) \cos(n\pi t).$$

Step 4. $f(t) = \frac{1}{2} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos((2k-1)\pi t)}{(2k-1)^2}.$

4.4 Applications

In this section, we will see how Fourier series can be used to find the exact value of certain infinite series.

Example 4.4.1

Use the Fourier series of the Sawtooth function in Example 4.3.8 to calculate the exact value of the infinite series

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots,$$

leaving the answer in terms of π .

Solution

From Example 4.3.8,

$$t = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nt) \quad \text{for } -\pi < t < \pi$$

Substituting $t = \frac{\pi}{2}$ into the Fourier series above, we have

$$\begin{aligned} \frac{\pi}{2} &= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi}{2}\right) \\ &= 2 \left(\frac{\sin \frac{\pi}{2}}{1} - \frac{\sin \pi}{2} + \frac{\sin \frac{3\pi}{2}}{3} - \frac{\sin 2\pi}{4} + \frac{\sin \frac{5\pi}{2}}{5} - \frac{\sin 3\pi}{6} + \dots \right) \\ &= 2 \left(1 - 0 + \frac{-1}{3} - 0 + \frac{1}{5} - 0 + \frac{-1}{7} - 0 + \dots \right) \end{aligned}$$

So,

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

4.5 Exercises

Part A: Baseline Problems

1. Sketch three cycles of the following periodic functions, and determine their periods (See Example 4.1.4).

$$(a) \quad f(t) = \begin{cases} -1 & \text{for } -\pi < t < 0 \\ 0 & \text{for } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi)$$

$$(b) \quad f(t) = \begin{cases} 2 & \text{for } -\pi < t < 0 \\ -2 & \text{for } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi)$$

$$(c) \quad f(t) = \begin{cases} 2 & \text{if } -1 < t < 0 \\ 1 & \text{if } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

$$(d) \quad f(t) = \begin{cases} -3 & \text{if } -1 < t < 0 \\ 2 & \text{if } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

- 2(a) to (d). Calculate the Fourier series of the functions in Problem 1(a) to (d), respectively (See Example 4.2.3).

Part B: Intermediate Problems

3. Sketch three cycles of the following periodic functions, and determine their periods (See Example 4.1.4).

$$(a) \quad f(t) = \begin{cases} -t & \text{for } -1 < t \leq 0 \\ 0 & \text{for } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

$$(b) \quad f(t) = \begin{cases} 0 & \text{for } -\pi < t \leq 0 \\ 2t & \text{for } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi)$$

$$(c) \quad f(t) = \begin{cases} 1 & \text{for } -1 < t < 0 \\ 1-t & \text{for } 0 \leq t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

$$(d) \quad f(t) = t + 1 \text{ for } -1 < t < 1 \quad \text{and} \quad f(t) = f(t + 2)$$

4(a) to (d). Calculate the Fourier series of the functions in Problem 2(a) to (d), respectively (See Example 4.2.4).

5. Sketch three cycles of the following periodic functions, and for each function, determine whether it is even or odd function (See Example 4.3.2).

$$(a) \quad f(t) = \begin{cases} -3 & \text{if } -1 < t < 0 \\ 3 & \text{if } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

$$(b) \quad f(t) = \begin{cases} t & \text{if } -\pi < t \leq 0 \\ -t & \text{if } 0 < t < \pi \end{cases} \quad \text{and} \quad f(t) = f(t + 2\pi)$$

$$(c) \quad f(t) = \begin{cases} 1 + t & \text{if } -1 < t \leq 0 \\ 1 - t & \text{if } 0 < t < 1 \end{cases} \quad \text{and} \quad f(t) = f(t + 2)$$

$$(d) \quad f(t) = -2t \text{ for } -\pi < t < \pi \quad \text{and} \quad f(t) = f(t + 2\pi)$$

6(a) to (d). Calculate the Fourier series for the functions in Problem 5(a) to (d), respectively (See Examples 4.3.7 – 4.3.9).

Part C: Application Problems

(See Section 4.4)

7. (a) Calculate the Fourier Series of $f(t)$ given by

$$f(t) = t - t^2 \text{ for } -\pi < t < \pi \quad \text{and} \quad f(t) = f(t + 2\pi)$$

(b) Hence, calculate the exact value of the infinite series

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \cdots,$$

leaving the answer in terms of π .

8. (a) Calculate the Fourier Series of $f(t)$ given by
 $f(t) = t \sin t$ for $-\pi < t < \pi$ and $f(t) = f(t + 2\pi)$
- (b) Hence, calculate the exact value of the infinite series

$$\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \frac{1}{7 \cdot 9} + \frac{1}{9 \cdot 11} - \dots,$$
leaving the answer in terms of π .

Part D: Beyond the Concepts

9. Use the Fourier series of $f(t)$ in Problem 5(b) to calculate the exact value of the following infinite series, leaving the answer in terms of π .
- (a) $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$
- (b) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ (rearrangement of terms is allowed here)
- (c) $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$ by using the **Parseval's Identity**

$$\frac{2}{T} \int_{-T/2}^{T/2} [f(t)]^2 dt = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Remark. The infinite series $\sum_{k=1}^{\infty} \frac{1}{k^s}$ is called the **Riemann Zeta function**.